Calculus & Statistics

Tutorial-7

Double integration & Area by double Integration

Q1. Evaluate the following integrals:

i. $\int_0^1 \int_1^2 (4x^3 - 9x^2y^2) dydx$

ii. $\int_1^4 \int_1^4 \left(\frac{x}{y} + \frac{y}{x}\right) dydx$

iii. $\int_{\pi/6}^{\pi/2} \int_{-1}^{5} cosy\, dxdy$

iv. $\int_0^1 \int_0^1 xy\sqrt{x^2 + y^2} dydx$

v. $\int_0^2 \int_0^\pi rsin^2\theta d\theta dr$

vi. $\int_0^4 \int_0^{\sqrt{y}} xy^2 dxdy$

vii. $\int_0^1 \int_{x^2}^{x} (1 + 2y) dydx$

viii. $\int_0^1 \int_{2x}^{2} (x - y) dydx$

ix. $\int_0^1 \int_0^v \sqrt{1 - v^2} dudv$

x. $\int_0^1 \int_0^{\sqrt{1+x^2}} \frac{dxdy}{1+x^2+y^2}$

Q2. Evaluate the following integrals over the given region:

i. Evaluate $\iint xy(x + y) \ \ dxdy \ \ over\ the\ area\ between\ y = x^2\ \ and\ \ y = x$

ii. Evaluate $\iint xydxdy$, over the region enclosed by the circle $x^2 + y^2 - 2x = 0$,the parabola $y^2 = 2x$ & the line $y = 2x$.

iii. Evaluate $\iint x^2 dxdy$, over the region in the first quadrant bounded by the lines $y = 0, x = 8, y = x$, and the curve $xy = 16$.

iv. Evaluate $\iint xydxdy$, over the region bounded by $x - axis$, the line $x = 2a$ & the parabola $x^2 = 4ay$

v. Evaluate $\int\int r^3 dr\, d\theta$ over the area bounded between the circles $r = 2cos\theta$ & $r = 4cos\theta$

Q3. Evaluate the following integrals by changing the order of integration.

i. $\int_0^{\pi}\int_x^{\pi}\frac{\sin y}{y}\,dydx$

ii. $\int_{-2}^{1}\int_{x^2+4x}^{3x+2}dydx$

iii. $\int_0^1\int_x^{\sqrt{(2-x^2)}}\frac{x\,dydx}{\sqrt{x^2+y^2}}$

iv. $\int_0^1\int_0^{\sqrt{(1-x^2)}}y^2\,dxdy$

v. $\int_0^{\infty}\int_0^{x}xe^{\frac{-x^2}{y}}\,dydx$

Q4. Evaluate the following integrals by changing to polar co-ordinates.

i. $\int_0^a\int_y^a\frac{x^2dydx}{\sqrt{x^2+y^2}}$

ii. $\int_0^{4a}\int_{\frac{y^2}{4a}}^{y}\frac{x^2-y^2}{x^2+y^2}\,dxdy$

iii. $\iint\sqrt{\frac{1-x^2-y^2}{1+x^2+y^2}}\,dxdy$ over the first quadrant of circle $x^2+y^2=1.$

iv. $\int_0^1\int_x^{\sqrt{(2-x^2)}}\frac{x}{\sqrt{x^2+y^2}}\,dydx$

v. $\int_0^{\frac{a}{\sqrt{2}}}\int_y^{\sqrt{(a^2-y^2)}}\log(x^2+y^2)\,dxdy$

Q5. Evaluate the following areas by double integration:

i. Find the area bounded by the parabola $y^2=4x$ and the line $2x-3y+4=0$

ii. Find the area between the circles $r=4\sin\theta$ and $r=6\sin\theta$.

iii. Find the area bounded by the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$,above x-axis.

iv. Find the area enclosed by the parabolas $y^2=4ax$ & $x^2=4ay$

v. Find the area enclosed by the parabola $y=4x-x^2$ and the line $y=x$